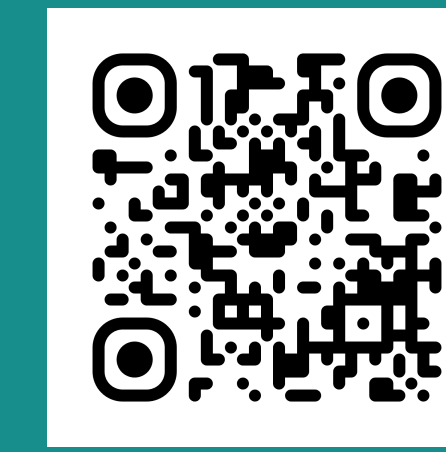


MedROI: Codec-Agnostic Region of Interest-Centric Compression for Medical Images

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Our code link



Introduction

Motivation

Medical imaging data is growing fast, but existing codecs still waste bits on diagnostically irrelevant background.

Key Contributions

- 1) **First to fully remove background** (not degrade it) before compression
- 2) Codec-agnostic across 6 representative codecs
- 3) Up to **81.3:1** compression, clinically acceptable quality

Methods

ROI Extraction

- Extract brain tissue using intensity thresholding (threshold = mean intensity of non-zero voxels)
- Compute a tight 3D bounding box enclosing tissue voxels
- Apply adaptive padding (+3 voxels) to preserve tissue.

Metadata

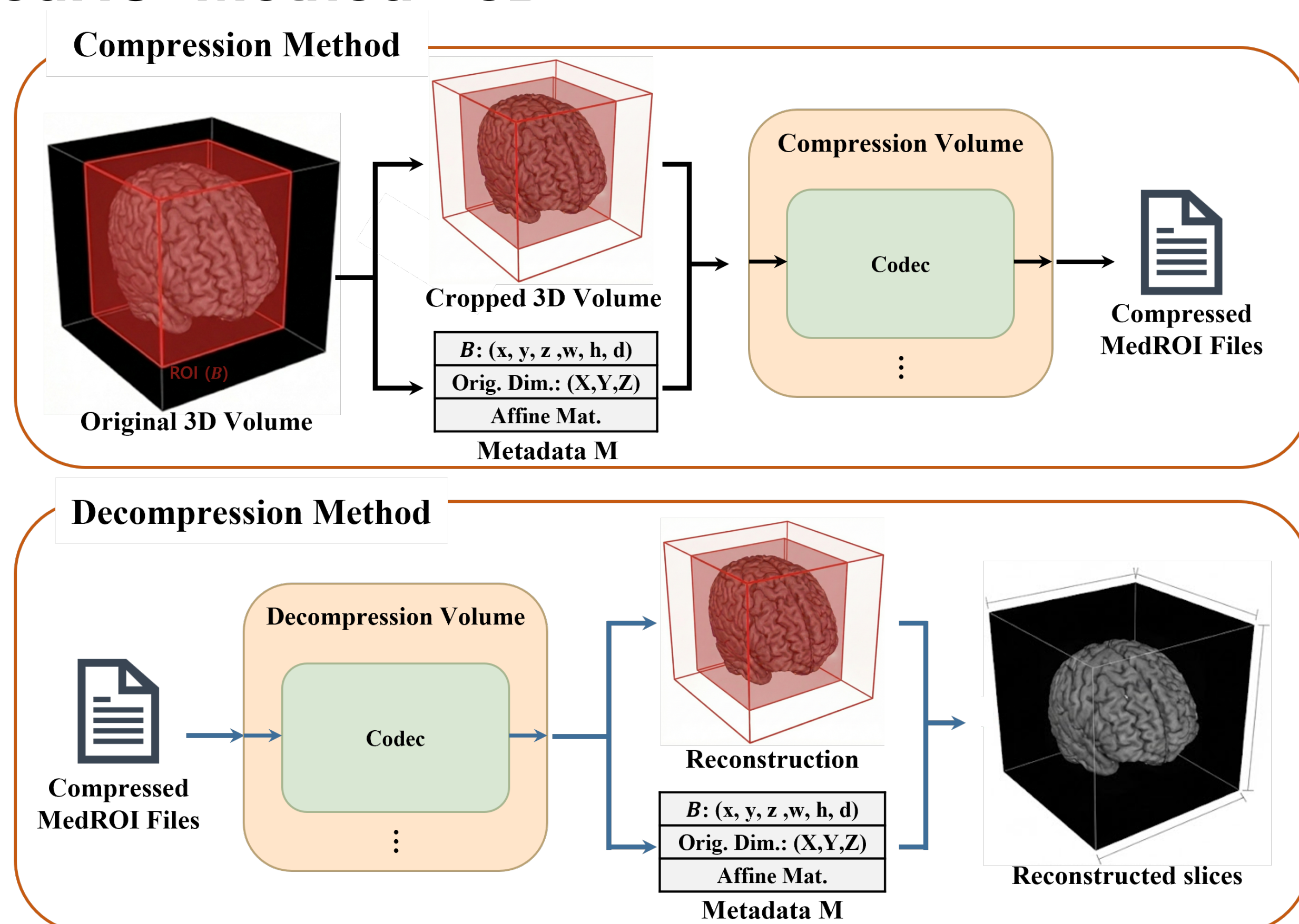
Store only 54 bytes for reconstruction

Component	Fields	Type	Size (bytes)
Bounding Box	$(x_{\min}, x_{\max}, y_{\min}, y_{\max}, z_{\min}, z_{\max})$	int16	12
Original Shape	(W, H, D)	int16	6
Affine Matrix	3×3 rotation-scaling submatrix	float32	36
Total	-	-	54

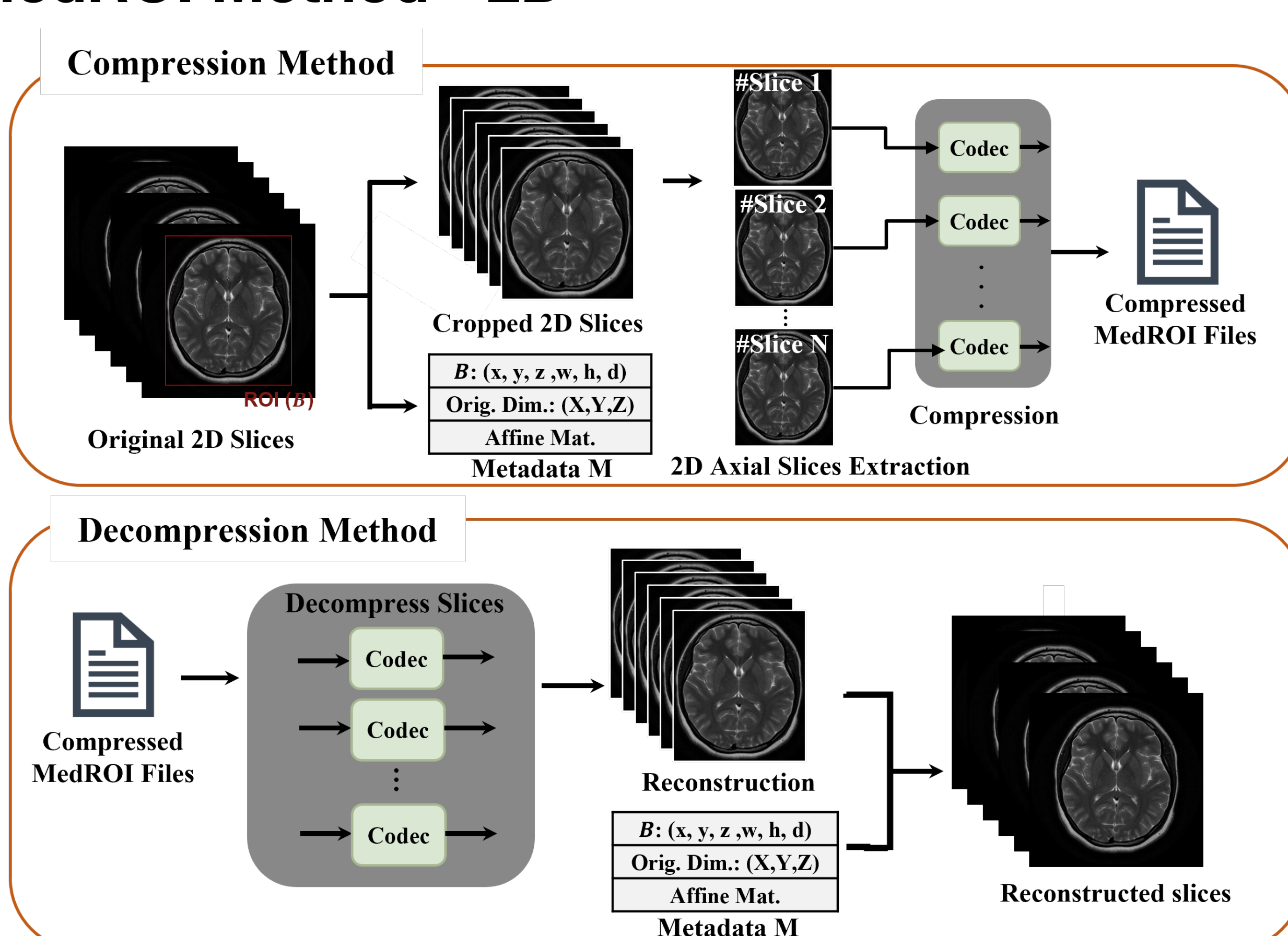
Dataset

ADNI dataset - 200 subjects

MedROI Method - 3D



MedROI Method - 2D



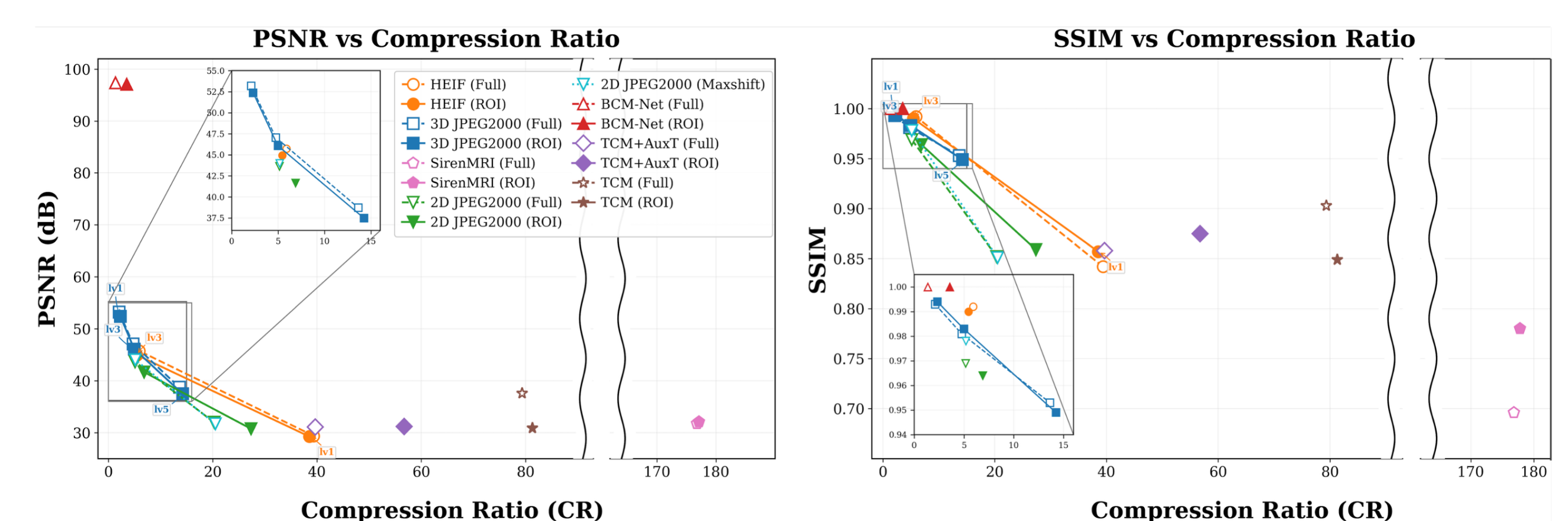
Result

Quantitative Result

Type	Model	Method	CR $\uparrow$	BPP $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	Comp./Decomp. (s)
2D	JPEG2000 lv3	Full	20.35	0.395	33.03	0.852	1.701 / 0.350
		Maxshift	20.51	0.390	31.70	0.851	2.492 / 1.994
		MedROI	27.37	0.397	30.73	0.859	1.380 / 0.265
	HEIF lv3	Full	5.907	1.924	45.74	0.992	0.112 / 2.9e-3
		MedROI	5.450	1.650	44.94	0.990	0.106 / 2.7e-3
	TCM+AuxT	Full	39.67	0.216	31.10	0.858	19.80 / 21.46
		MedROI	56.75	0.144	31.18	0.875	18.00 / 19.46
	TCM	Full	79.37	0.105	37.59	0.903	20.56 / 21.41
		MedROI	81.31	0.102	30.89	0.849	17.84 / 18.60
	BCM-Net	Full	1.35	11.96	97.34	1.000	410.0 / 15.06
		MedROI	3.55	9.313	97.07	1.000	388.2 / 9.062
3D	JPEG2000 lv5	Full	13.63	0.615	38.72	0.953	2.742 / 1.009
		MedROI	14.23	0.591	37.49	0.949	2.446 / 0.204
	SirenMRI	Full	176.8	0.365	31.74	0.696	720.7 / 1.000
		MedROI	177.8	0.364	32.08	0.780	540.6 / 1.310

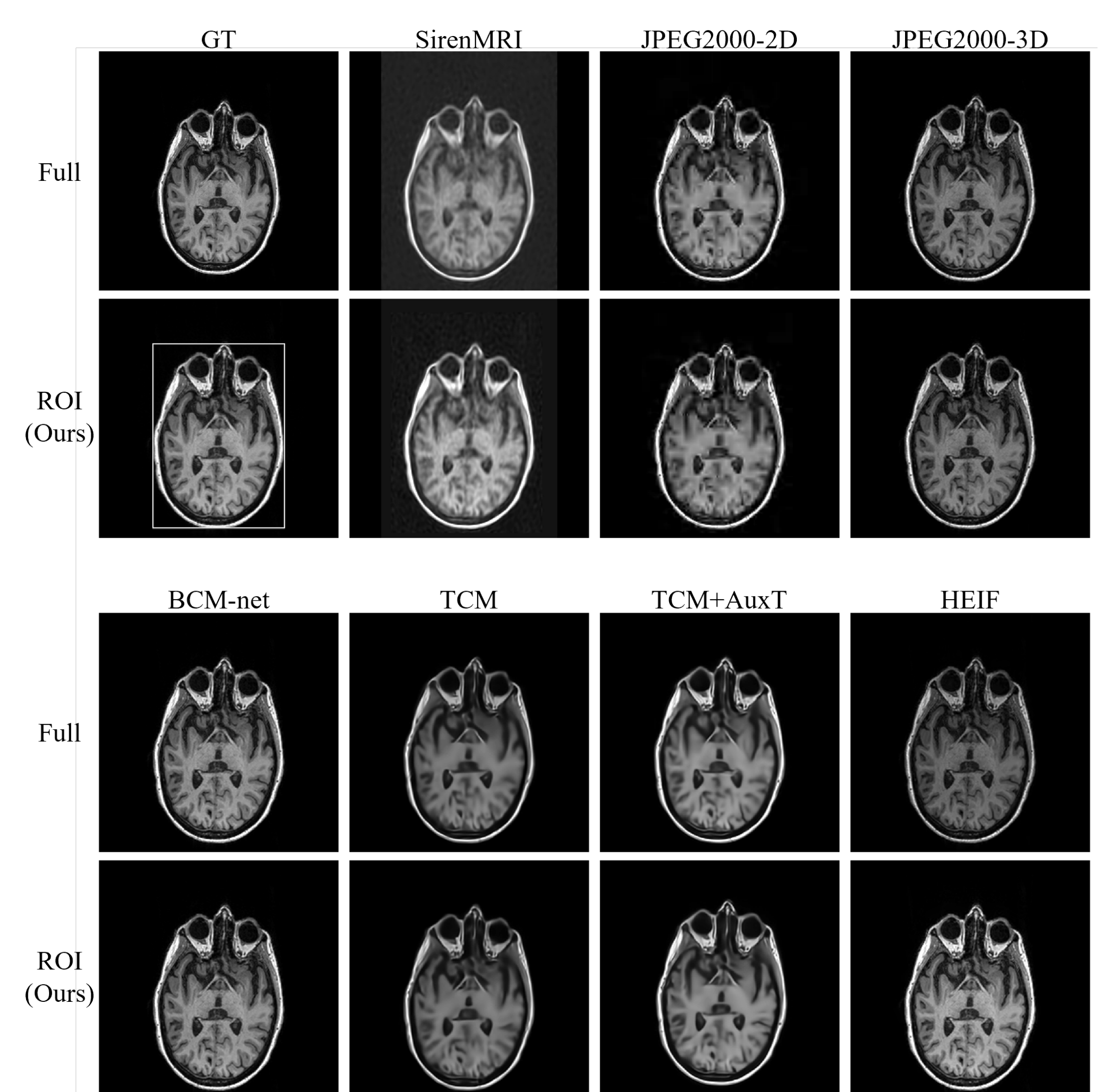
- MedROI improves CR and speed across most codecs, with comparable PSNR/SSIM within the ROI.

Rate-Distortion Analysis



- ROI curves (filled) shift rightward vs. full-volume (hollow), showing higher CR at similar quality.

Qualitative Comparison



- Full and ROI reconstructions show visually comparable quality across all codecs.

Conclusion

Conclusion & Future work

- MedROI: first framework to **fully remove** background before compression — codec-agnostic, plug-and-play
- Consistent gains in **compression ratio & speed** with comparable quality across 6 codecs
- **Future work:** extend to other modalities/anatomies, refine ROI extraction beyond thresholding